Python | Bilateral Filtering

Last Updated : 04 Jan, 2023

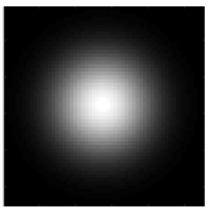
A bilateral filter is used for smoothening images and reducing noise, while **preserving edges**. This article explains an approach using the averaging filter, while this article provides one using a median filter. However, these convolutions often result in a loss of important edge information, since they blur out everything, irrespective of it being noise or an edge. To counter this problem, the non-linear bilateral filter was introduced.

Gaussian Blur

Gaussian blurring can be formulated as follows:

$$GB[I]_p = \sum_{q \epsilon S} G_\sigma(||p-q||) I_q$$

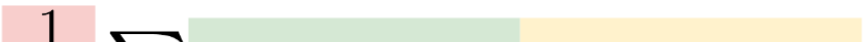
Normalized Gaussian Function



Here, $GA[I]_p$ is the result at pixel *p*, and the RHS is essentially a sum over all pixels *q* weighted by the Gaussian function. I_q is the intensity at pixel *q*.

Bilateral Filter: an Additional Edge Term

The bilateral filter can be formulated as follows:

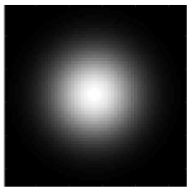


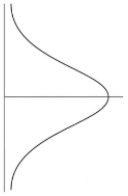
$$BF[I]_p = \frac{1}{W_p} \sum_{q \epsilon S} G_{\sigma_s}(||p-q||) G_{\sigma_r}(|I_p - I_q|) I_q$$

Normalization Factor

Space Weight

Range Weight



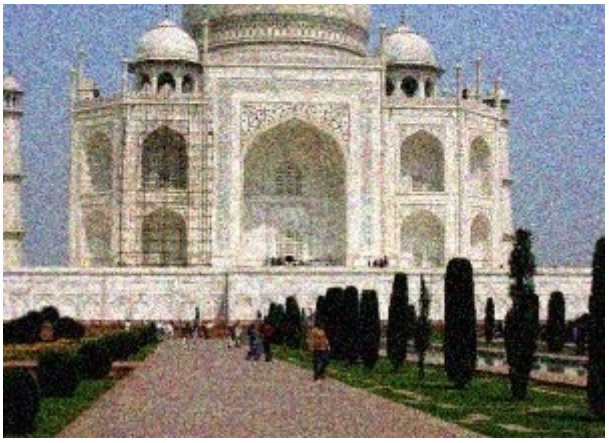


Here, the normalization factor and the range weight are new terms added to the previous equation. σ_s denotes the spatial extent of the kernel, i.e. the size of the neighborhood, and σ_r denotes the minimum amplitude of an edge. It ensures that only those pixels with intensity values similar to that of the central pixel are considered for blurring, while sharp intensity changes are maintained. The smaller the value of σ_r, the sharper the edge. As σ_r tends to infinity, the equation tends to a Gaussian blur.
OpenCV has a function called **bilateralFilter()** with the following arguments:

1. **d:** Diameter of each pixel neighborhood.
2. **sigmaColor:** Value of σ in the color space. The greater the value, the colors farther to each other will start to get mixed.
3. **sigmaSpace:** Value of σ in the coordinate space. The greater its value, the more further pixels will mix together, given that their colors lie within the sigmaColor range.

Code :

Input : Noisy Image.



Code : Implementing Bilateral Filtering

```
import cv2

# Read the image.
img = cv2.imread('taj.jpg')

# Apply bilateral filter with d = 15,
# sigmaColor = sigmaSpace = 75.
bilateral = cv2.bilateralFilter(img, 15, 75, 75)

# Save the output.
cv2.imwrite('taj_bilateral.jpg', bilateral)
```

Output of Bilateral Filter

Comparison with Average and Median filters

Below is the output of the average filter (cv2.blur(img, (5, 5))).

Below is the output of the median filter (cv2.medianBlur(img, 5)).

Below is the output of the Gaussian filter (cv2.GaussianBlur(img, (5, 5), 0)).

It is easy to note that all these denoising filters smudge the edges, while Bilateral Filtering retains them.